烧录下载

低功耗外部RTC
+纽扣电池

CH397
USB 转网口

UART

I2C

USB

电源模块

5V/3.3V

ESP32-S3

USB HUB

CH334P

USB

DOWNGO

5V/4V

UART

UART

ML307R-DL

蓝牙MESH

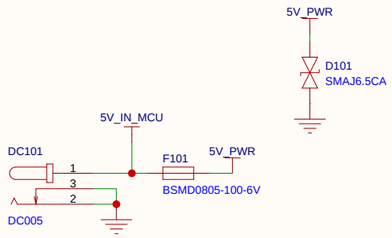
485接口

USB

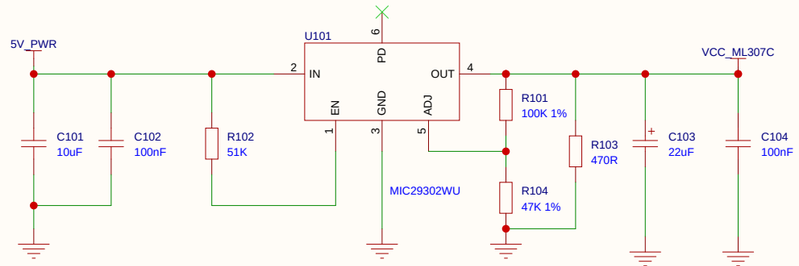
1. 主控采用ESP32-S3-WROOM，通过4G模块的ML307C进行数据上报，同时也支持沁恒的CH397以太网通讯
2. 输入5V的直流电源，3.3V降压电路和4V 降压电路，电流要求达到2A
3. 传感器采用RS485的自动收发进行通讯，1路485
4. 外挂蓝牙mesh芯片8250,通过串口通讯
5. 外接usb hub的芯片，实现1个USB接口可以同事通讯4个USB设备

更新说明：
1. 将2PIN的电源端更换为TX和RX

5V稳压电源输入

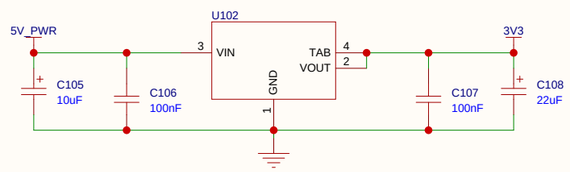


LDO 5V->3.8V

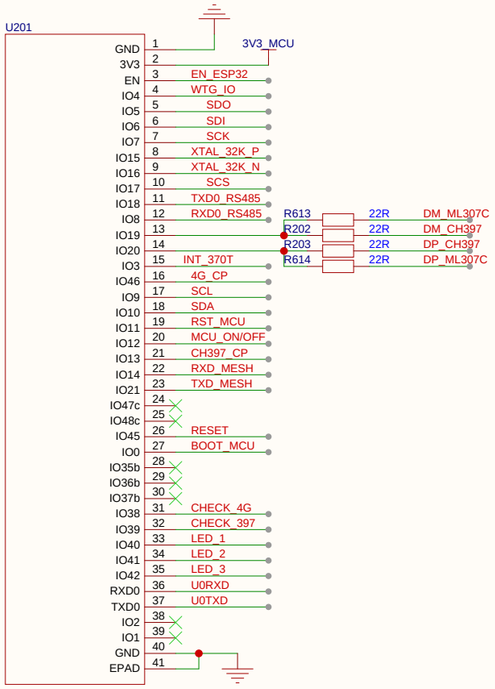


Vout = 1.242 * （R25 / R26 + 1）= 3.88V

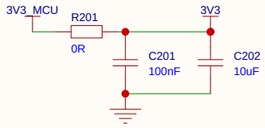
LDO 5V->3.3V

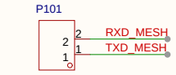


MCU

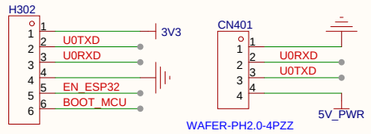


POWER

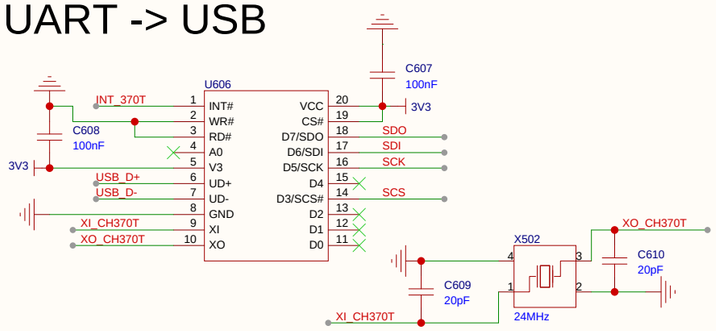




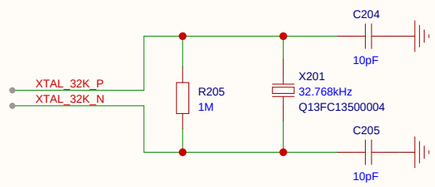
烧录接口



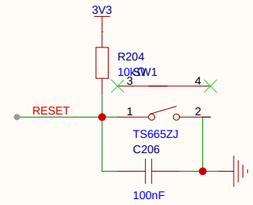
UART -> USB



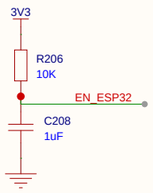
RTC



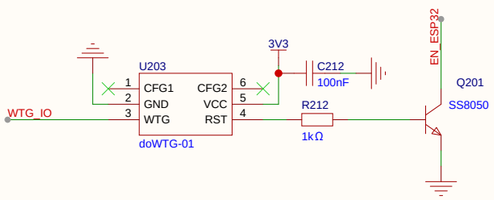
配置RESET



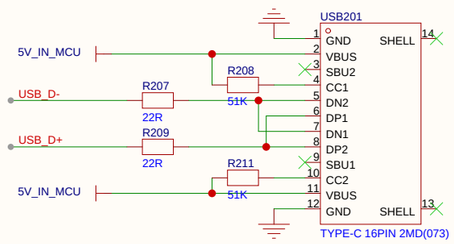
RESET



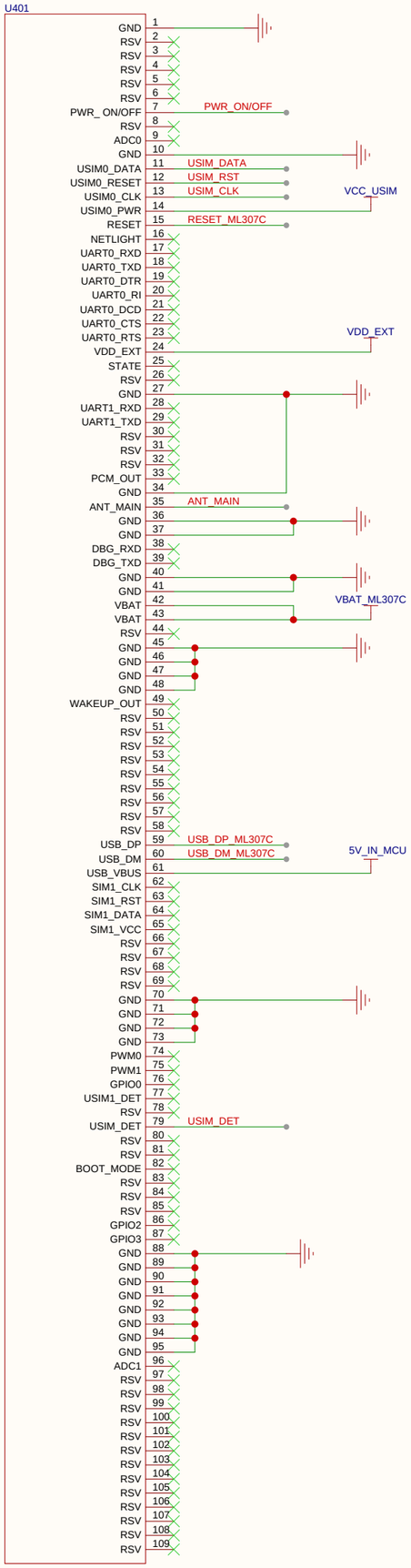
WATCH DOG

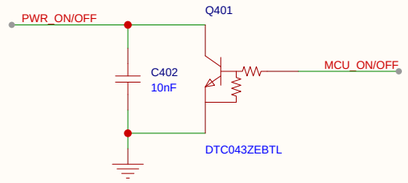


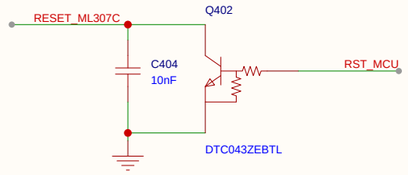
USB_DOWNGO

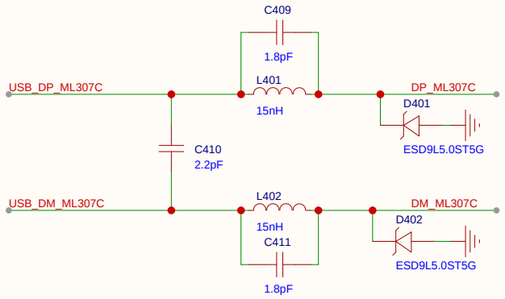


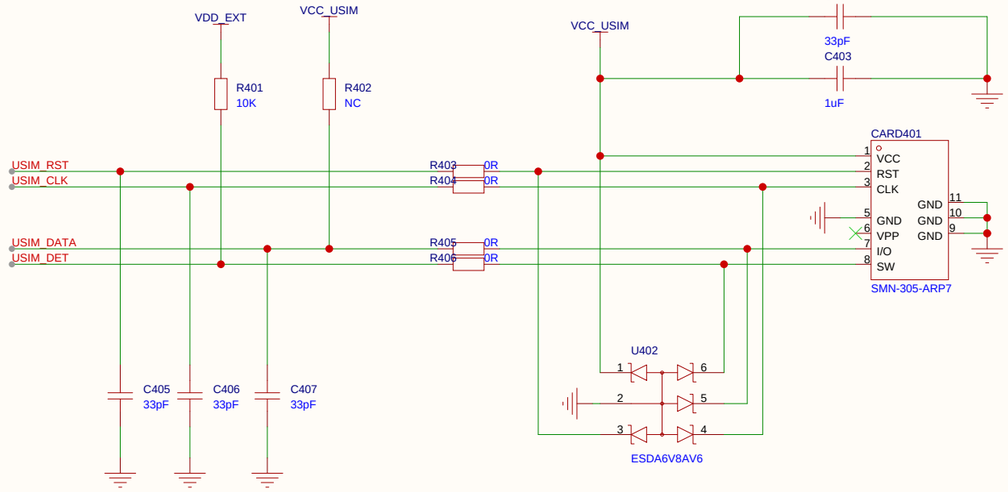
MCU

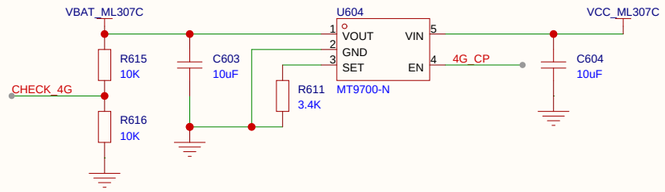

U401
GND 1
RSV 2
RSV 3
RSV 4
RSV 5
RSV 6
PWR_ ON/OFF 7 PWR_ON/OFF
RSV 8
ADC0 9
GND 10
USIM0_DATA 11 USIM_DATA
USIM0_RESET 12 USIM_RST
USIM0_CLK 13 USIM_CLK
USIM0_PWR 14 VCC_USIM
RESET 15 RESET_ML307C
NETLIGHT 16
UART0_RXD 17
UART0_TXD 18
UART0_DTR 19
UART0_RI 20
UART0_DCD 21
UART0_CTS 22
UART0_RTS 23
VDD_EXT 24 VDD_EXT
STATE 25
RSV 26
GND 27
UART1_RXD 28
UART1_TXD 29
RSV 30
RSV 31
RSV 32
PCM_OUT 33
GND 34
ANT_MAIN 35 ANT_MAIN
GND 36
GND 37
DBG_RXD 38
DBG_TXD 39
GND 40
GND 41
VBAT 42 VBAT_ML307C
VBAT 43
RSV 44
GND 45
GND 46
GND 47
GND 48
WAKEUP_OUT 49
RSV 50
RSV 51
RSV 52
RSV 53
RSV 54
RSV 55
RSV 56
RSV 57
RSV 58
USB_DP 59 USB_DP_ML307C
USB_DM 60 USB_DM_ML307C
USB_VBUS 61 5V_IN_MCU
SIM1_CLK 62
SIM1_RST 63
SIM1_DATA 64
SIM1_VCC 65
RSV 66
RSV 67
RSV 68
RSV 69
GND 70
GND 71
GND 72
GND 73
PWM0 74
PWM1 75
GPIO0 76
USIM1_DET 77
RSV 78
USIM_DET 79 USIM_DET
RSV 80
RSV 81
BOOT_MODE 82
RSV 83
RSV 84
RSV 85
GPIO2 86
GPIO3 87
GND 88
GND 89
GND 90
GND 91
GND 92
GND 93
GND 94
GND 95
ADC1 96
RSV 97
RSV 98
RSV 99
RSV 100
RSV 101
RSV 102
RSV 103
RSV 104
RSV 105
RSV 106
RSV 107
RSV 108
RSV 109

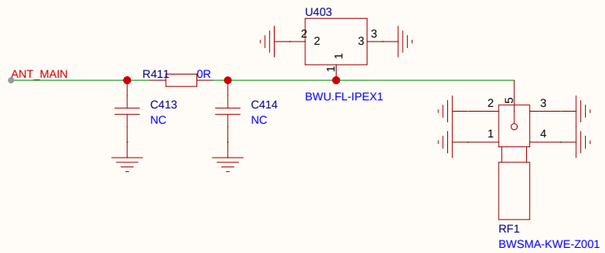

PWR_ON/OFF
Q401
C402
10nF
MCU_ON/OFF
DTC043ZEBTL


RESET_ML307C
Q402
C404
10nF
RST_MCU
DTC043ZEBTL


C409
1.8pF
USB_DP_ML307C
L401
15nH
DP_ML307C
D401
ESD9L5.0ST5G
C410
2.2pF
USB_DM_ML307C
L402
15nH
C411
1.8pF
DM_ML307C
D402
ESD9L5.0ST5G

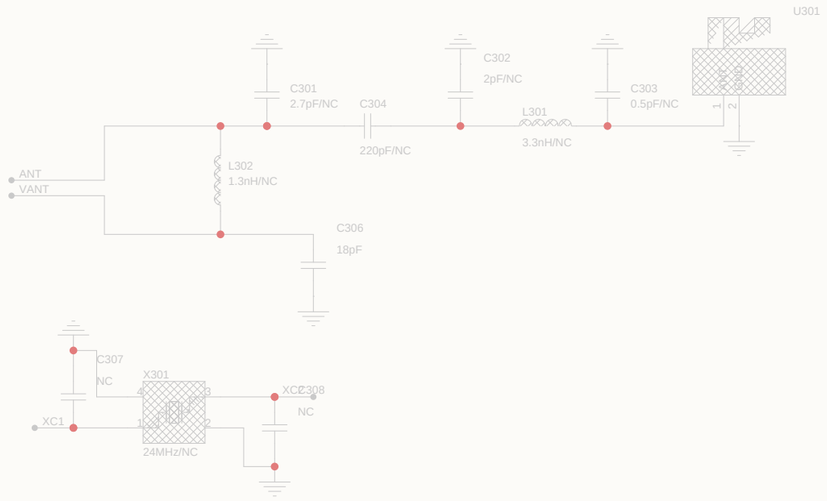

VDD_EXT
VCC_USIM
R401
10K
R402
NC
USIM_RST
USIM_CLK
USIM_DATA
USIM_DET
R403 0R
R404 0R
R405 0R
R406 0R
C405
33pF
C406
33pF
C407
33pF
VCC_USIM
C401
33pF
C403
1uF
CARD401
VCC
RST
CLK
GND
VPP
I/O
SW
GND
GND
GND
SMN-305-ARP7
U402
ESDA6V8AV6

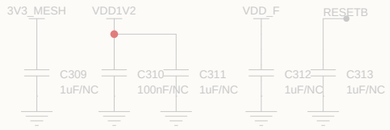

VBAT_ML307C
R615
10K
CHECK_4G
R616
10K
C603
10uF
R611
3.4K
U604
VOUT
GND
SET
VIN
EN
4G_CP
MT9700-N
VCC_ML307C
C604
10uF


U403
ANT_MAIN
R411 0R
C413
NC
C414
NC
BWU.FL-IPEX1
RF1
BWSMA-KWE-Z001

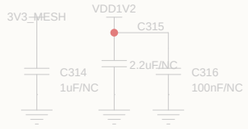
TELINK-8250


U302
1
2 TXD_MESH
3 RXD_MESH
4
5 SWS
6
7
8 C305 1uF/NC
9 3V3_MESH
10 L303 47uH/NC
11
12
13 VDD1V2
VDD_F
14
15
16
17
18
19 3V3_MESH
20
21
22
23
24
25 RESETB
26 VANT
27 ANT
28 VDD1V2
29 XC1
30 XC2
31
32
33


U301
C301
2.7pF/NC
C304
220pF/NC
C302
2pF/NC
L301
3.3nH/NC
C303
0.5pF/NC
ANT
VANT
L302
1.3nH/NC
C306
18pF
C307
NC
X301
24MHz/NC
XC1
XC2
C308
NC

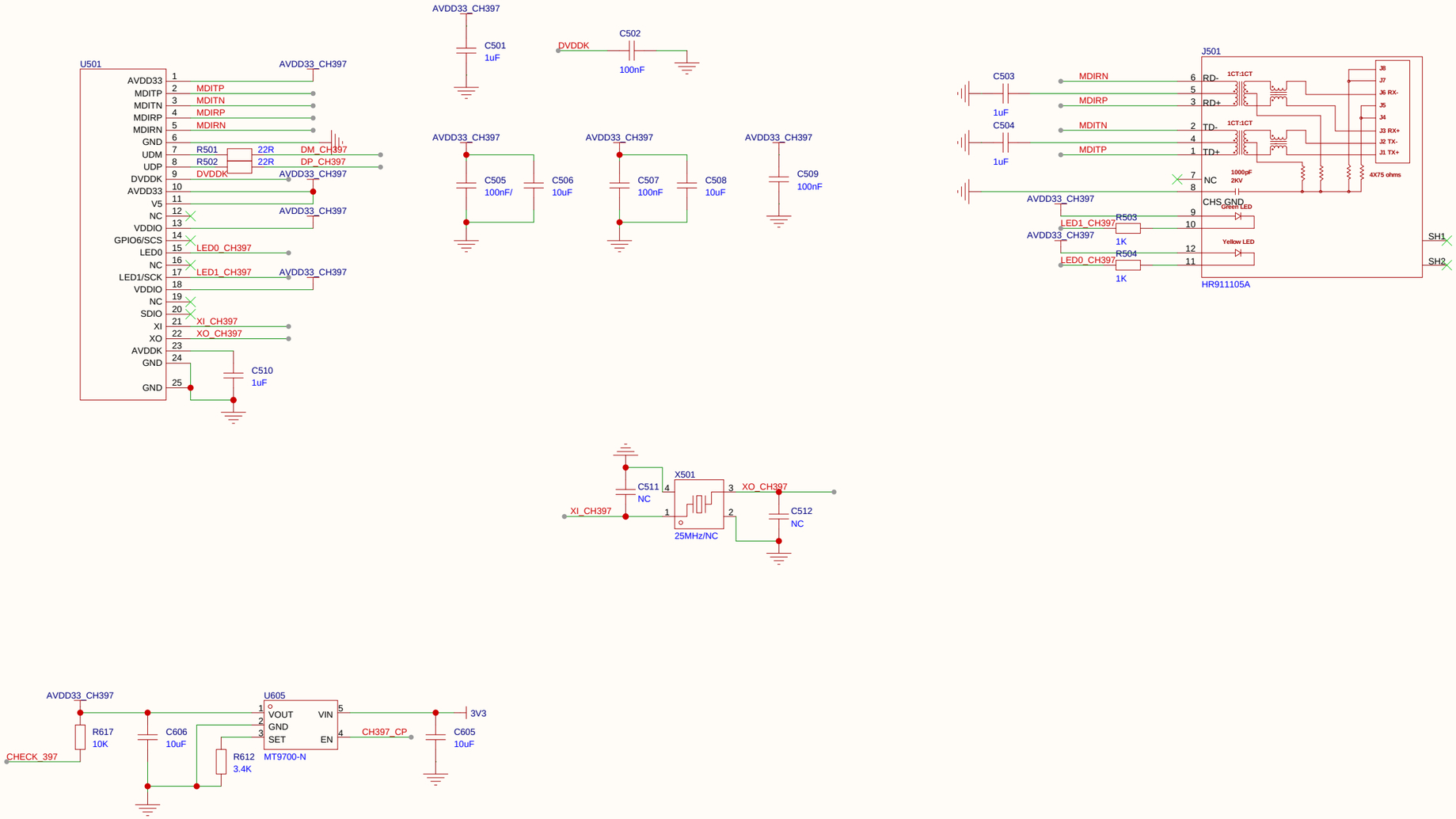

3V3_MESH
VDD1V2
VDD_F
RESETB
C309
1uF/NC
C310
100nF/NC
C311
1uF/NC
C312
1uF/NC
C313
1uF/NC

3V3_MESH
L304
3V3
GZ1608D601TF/NC

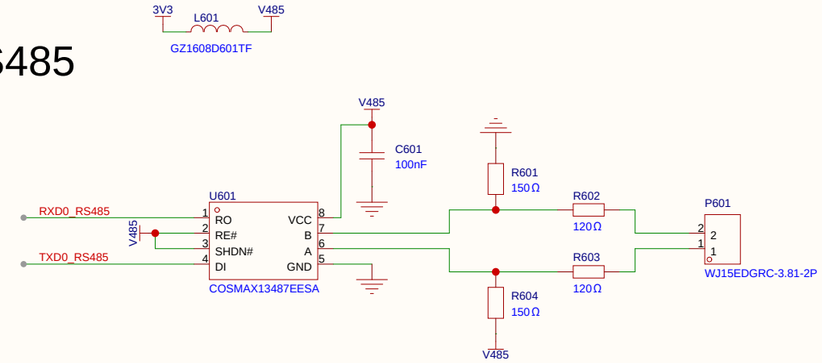

3V3_MESH
VDD1V2
C315
C314
1uF/NC
2.2uF/NC
C316
100nF/NC


H301
1
2
3
3V3
SWS

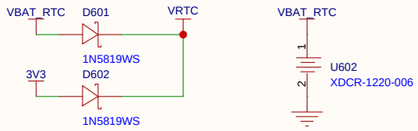
3V3_MESH
C317
1uF/NC

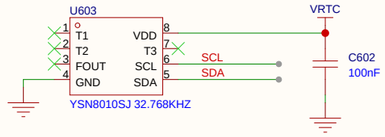

U501
AVDD33
MDITP
MDITN
MDIRP
MDIRN
GND
UDM
UDP
DVDDK
AVDD33
V5
NC
VDDIO
GPIO6/SCS
LED0
NC
LED1/SCK
VDDIO
NC
SDIO
XI
XO
AVDDK
GND
GND
AVDD33_CH397
MDITP
MDITN
MDIRP
MDIRN
R501
22R
DM_CH397
R502
22R
DP_CH397
DVDDK
AVDD33_CH397
AVDD33_CH397
LED0_CH397
LED1_CH397
AVDD33_CH397
XI_CH397
XO_CH397
C510
1uF
AVDD33_CH397
C501
1uF
DVDDK
C502
100nF
AVDD33_CH397
C505
100nF/
C506
10uF
AVDD33_CH397
C507
100nF
C508
10uF
AVDD33_CH397
C509
100nF
X501
C511
NC
XI_CH397
1
4
3
XO_CH397
2
C512
NC
25MHz/NC
J501
C503
1uF
C504
1uF
MDIRN
MDIRP
MDITN
MDITP
RD-
RD+
TD-
TD+
NC
CHS_GND
1CT:1CT
1CT:1CT
1000pF
2KV
AVDD33_CH397
LED1_CH397
R503
1K
AVDD33_CH397
LED0_CH397
R504
1K
Green LED
Yellow LED
J8
J7
J6 RX-
J5
J4
J3 RX+
J2 TX-
J1 TX+
4X75 ohms
SH1
SH2
HR911105A
AVDD33_CH397
R617
10K
CHECK_397
C606
10uF
U605
VOUT
GND
SET
VIN
EN
MT9700-N
R612
3.4K
CH397_CP
3V3
C605
10uF

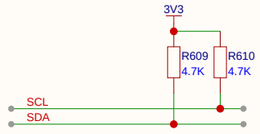
RS485

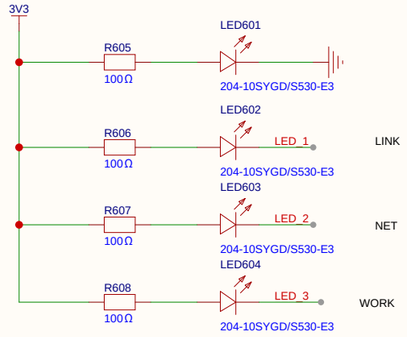

3V3
L601
V485
GZ1608D601TF
V485
C601
100nF
U601
RXD0_RS485
TXD0_RS485
V485
RO
RE#
SHDN#
DI
VCC
B
A
GND
COSMAX13487EESA
R601
150Ω
R602
120Ω
R603
120Ω
R604
150Ω
V485
P601
WJ15EDGRC-3.81-2P

RTC


VBAT_RTC
D601
VRTC
1N5819WS
3V3
D602
1N5819WS
VBAT_RTC
U602
XDCR-1220-006


U603
T1
T2
FOUT
GND
VDD
T3
SCL
SDA
YSN8010SJ 32.768KHZ
VRTC
C602
100nF
SCL
SDA


3V3
R609
4.7K
R610
4.7K
SCL
SDA


3V3
R605
100Ω
LED601
204-10SYGD/S530-E3
R606
100Ω
LED602
LED_1
204-10SYGD/S530-E3
LINK
R607
100Ω
LED603
LED_2
204-10SYGD/S530-E3
NET
R608
100Ω
LED604
LED_3
204-10SYGD/S530-E3
WORK